

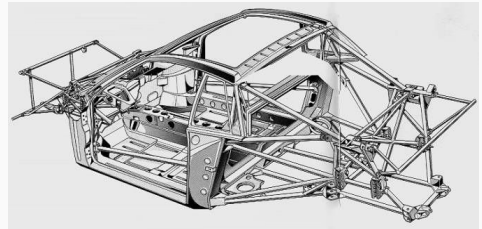
Dynamic analysis of frames and grids

Special structural elements

ME473 Dynamic finite element analysis of structures

Stefano Burzio

2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin method	
3		Finite element method	Groups formation
4		Systematization of the procedure	Project 1 statement
5		3d elements, numerical integration	
6	Special structural elements	Bars and trusses	
7		Planar beams	Project 1 submission
8		Frames and grids	Project 2 statement

Summary

- Recap week 7
- Plane frames
- Plane grids
- Three-dimensional frames

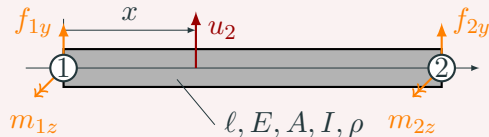
Recommended readings

- ① Logan, A first course in the finite element method, 6th ed. (chap. 5)
- ② Paz and Leigh, Structural dynamics, 6th ed. (chap. 11, 12 and 13)
- ③ Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis, 2nd ed. (chap. 7, 8 and 9)

Recap week 7

Vibrations of planar beams

Planar thin beam element



Equation governing the dynamics:

$$\partial_{xx}^2 (EI \partial_{xx}^2 u_2(x, t)) + \rho A \ddot{u}_2(x, t) = 0$$

■ Displacement approximation:

$$u_2^h(x, t) = \mathbf{H}(x) \mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x) & h_2(x) & h_3(x) & h_4(x) \end{bmatrix} \begin{bmatrix} q_{1y}(t) \\ \phi_{1z}(t) \\ q_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}$$

■ Hermite local shape functions:

$$h_1(x) = 2(x/\ell)^3 - 3(x/\ell)^2 + 1$$

$$h_2(x) = x(1 - x/\ell)^2$$

$$h_3(x) = 3(x/\ell)^2 - 2(x/\ell)$$

$$h_4(x) = x(x/\ell)(x/\ell - 1)$$

Discretization of thin beams

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell EI \frac{d^2 \mathbf{H}^T}{dx^2} \frac{d^2 \mathbf{H}}{dx^2} dx = \frac{EI}{\ell^3} \begin{bmatrix} 12 & 6\ell & -12 & 6\ell \\ & 4\ell^2 & -6\ell & 2\ell^2 \\ & & 12 & -6\ell \\ sym. & & & 4\ell^2 \end{bmatrix}$$

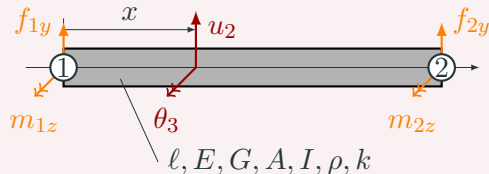
- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \int_0^\ell \rho A \mathbf{H}^T \mathbf{H} dx = \frac{\rho A \ell}{420} \begin{bmatrix} 156 & 22\ell & 54 & -13\ell \\ & 4\ell^2 & 13\ell & -3\ell^2 \\ & & 156 & -22\ell \\ sym. & & & 4\ell^2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} f_{1y}(t) \\ m_{1z}(t) \\ f_{2y}(t) \\ m_{2z}(t) \end{bmatrix}$$

Planar thick beam element



Equations governing the dynamics:

$$\partial_x(kGA(\partial_x u_2 - \theta_3)) + p = \rho A \ddot{u}_2$$

$$\partial_x(EI \partial_x \theta_3) + kGA(\partial_x u_2 - \theta_3) = \rho I \ddot{\theta}_3$$

- Displacement approximation $\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x})\mathbf{q}(t)$:

$$\begin{bmatrix} u_2^h(t, x) \\ \theta_3^h(t, x) \end{bmatrix} = \begin{bmatrix} h_1^c & h_2^c(x) & h_3^c(x) & h_4^c(x) \\ 0 & h_1^l(x) & 0 & h_2^l(x) \end{bmatrix} \begin{bmatrix} q'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

- h_i^c **Cubic Hermite shape functions** h_i^c are used to approximate the transversal displacement while **linear shape functions** h_j^l for the rotation.

Discretization of thick beams

- $\mathbf{B} = \nabla_u \mathbf{H}$ is the deformation matrix:

$$\mathbf{B} = \begin{bmatrix} \partial_x h_1^c & \partial_x h_2^c - h_1^l & \partial_x h_3^c & \partial_x h_4^c - h_2^l \\ 0 & \partial_x h_1^l & 0 & \partial_x h_2^l \end{bmatrix}$$

- The stiffness $\mathbf{K}$ and mass $\mathbf{Ma}$ matrices and load vector $\mathbf{r}$ are defined as

$$\mathbf{K} = \int_0^\ell \mathbf{B}^T \mathbf{C} \mathbf{B} \, dx, \quad \mathbf{Ma} = \int_0^\ell \mathbf{H}^T \mathbf{M} \mathbf{H} \, dx, \quad \mathbf{r}(t) = \int_0^\ell \mathbf{H}^T \mathbf{f} \, dx + \mathbf{H}^T(\ell) \hat{\mathbf{f}}.$$

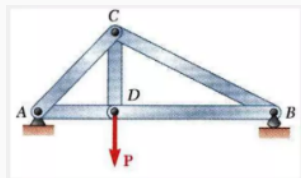
- Material constitutive matrices:

$$\mathbf{C} = \begin{bmatrix} kGA & 0 \\ 0 & EI \end{bmatrix} \quad \text{and} \quad \mathbf{M} = \begin{bmatrix} \rho A & 0 \\ 0 & \rho I \end{bmatrix}$$

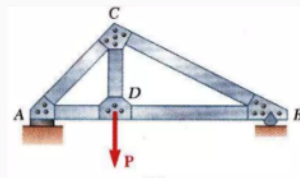
Plane frames

What is a plane frame?

- Structure composed of oriented thin beam elements, connected by welding, and carrying **transversal and axial forces** that all lies in a common plane.
- Both forces and moments can be transmitted between members.
- Loads are acting only in the common plane of the structure and they must be applied at the joints.



Plane truss

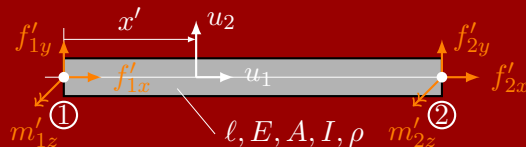


Plane frame

Truss + Thin beams = Frame

Axial loads in thin beam element

The inclusion of axial forces in a flexural beam element requires a *superposition* of bar and beam elements:



Differential equations governing the dynamics:

$$\partial_{x'x'}^2 (EI \partial_{x'x'}^2 u_2(x', t)) + \rho A \ddot{u}_2(x', t) = 0$$

$$EA \partial_{x'x'}^2 u_1(x', t) = \rho A \ddot{u}_1(x', t)$$

Axial effects are subsequently incorporated into the beam element formulation, unless specified otherwise.

Axial displacement in thin beam element

A beam element now has three degrees of freedom per node: q'_{ix} , q'_{iy} , and ϕ'_{iz} .

- Displacements approximation:

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t) = [h_1(x') \ h_2(x') \ h_3(x') \ h_4(x') \ h_5(x') \ h_6(x')] \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_3(x') = x'(1 - x'/\ell)^2$$

$$h_4(x') = x'/\ell$$

$$h_5(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_6(x') = x'(x'/\ell)(x'/\ell - 1)$$

Discretization of thin beam including axial effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3} \begin{bmatrix} A\ell^2/I & 0 & 0 & -A\ell^2/I & 0 & 0 \\ & 12 & 6\ell & 0 & -12 & 6\ell \\ & & 4\ell^2 & 0 & -6\ell & 2\ell^2 \\ & & & A\ell^2/I & 0 & 0 \\ & & & & 12 & -6\ell \\ sym. & & & & & 4\ell^2 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140 & 0 & 0 & 70 & 0 & 0 \\ & 0 & 156 & 22\ell & 0 & 54 \\ & & & 4\ell^2 & 0 & 13\ell \\ & & & & 70 & 0 \\ & & & & & 0 \\ & & & & & 156 \\ & & & & & & -22\ell \\ & & & & & & & 4\ell^2 \end{bmatrix}$$

Arbitrarily oriented thin beam element including axial effects

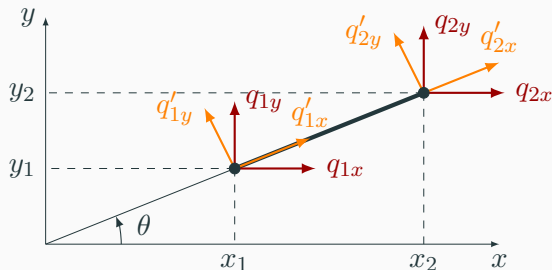
- Displacements in local coordinates:

$$\mathbf{q}_{loc} = [q'_{1x} \quad q'_{1y} \quad \phi'_{1z} \quad q'_{2x} \quad q'_{2y} \quad \phi'_{2z}]^T.$$

- Displacements in global coordinates:

$$\mathbf{q} = [q_{1x} \quad q_{1y} \quad \phi_{1z} \quad q_{2x} \quad q_{2y} \quad \phi_{2z}]^T.$$

- Relation between local and global displacements:



$$\underbrace{\begin{bmatrix} q'_{1x} \\ q'_{1y} \\ \phi'_{1z} \\ q'_{2x} \\ q'_{2y} \\ \phi'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ \phi_{1z} \\ q_{2x} \\ q_{2y} \\ \phi_{2z} \end{bmatrix}}_{\mathbf{q}}$$

Discretization of arbitrarily oriented beam including axial effects

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} = \frac{E}{\ell} \begin{bmatrix} \frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(-12I + A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} & -\frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(12I - A\ell^2)CS}{\ell^2} & \frac{-6IS}{\ell} \\ & \frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{6IC}{\ell} & \frac{(12I - A\ell^2)CS}{\ell^2} & -\frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{6IC}{\ell} \\ & & 4I & \frac{6IS}{\ell} & \frac{-6IC}{\ell} & 2I \\ & & & \frac{A\ell^3 C^2 + 12IS^2}{\ell^2} & \frac{(-12I + A\ell^2)CS}{\ell^2} & \frac{6IS}{\ell} \\ & & & & \frac{12IC^2 + A\ell^2 S^2}{\ell^2} & \frac{-6IC}{\ell} \\ & & & & & 4I \end{bmatrix}$$

where $C = \cos(\theta)$ and $S = \sin(\theta)$.

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

Applied loads

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} f'_{1x}(t) \\ f'_{1y}(t) \\ m'_{1z}(t) \\ f'_{2x}(t) \\ f'_{2y}(t) \\ m'_{2z}(t) \end{bmatrix}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc} = \begin{bmatrix} \cos(\theta)f'_{1x} + \sin(\theta)f'_{1y} \\ -\sin(\theta)f'_{1x} + \cos(\theta)f'_{1y} \\ m'_{1z} \\ \cos(\theta)f'_{2x} + \sin(\theta)f'_{2y} \\ -\sin(\theta)f'_{2x} + \cos(\theta)f'_{2y} \\ m'_{2z} \end{bmatrix}$$

Assembly of stiffness and mass matrices and loads vector

Given a 2d frame structure made of m oriented beams, n nodes, and 3 DOFs per node:

1. Element quantities:

- For each beam e , compute the element quantities global coordinates:

$${}^e\mathbf{K} = {}^e\mathbf{T}^T {}^e\mathbf{K}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{M} = {}^e\mathbf{T}^T {}^e\mathbf{M}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{f} = {}^e\mathbf{T}^T {}^e\mathbf{f}_{loc}$$

2. Global assembly:

- Initialize global stiffness matrix $\mathbf{K}$ and global mass matrix $\mathbf{M}$ of size $3n \times 3n$,
- Initialize global loads vector $\mathbf{f}$ of size $3n \times 1$,
- Assemble each ${}^e\mathbf{K}$, ${}^e\mathbf{M}$ and ${}^e\mathbf{f}$ for $e = 1, \dots, m$, into $\mathbf{K}$, $\mathbf{M}$ and $\mathbf{f}$ respectively using element connectivity.

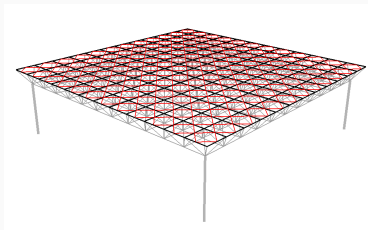
MATLAB example - a 2d frame in free vibration

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Plane grids

What is a grid?

- A grid is a structure composed of oriented planar beams subjected to **perpendicular loading** that produced significant bending effects.
- Beams are connected by welding: both forces and moments can be transmitted between members.
- Axial effect of axial displacement is ignored for the moment.
- Common type of structures used in to model floors, roofs, bridge decks, etc..



Degrees of freedom identification

■ Planar frame under action of loads in the plane of the structure:

Components required to describe the displacements of joint i are:

- q_{ix} translation in the x direction,
- q_{iy} translation in the y direction,
- ϕ_{iz} bending, rotation about the z axis.

■ Planar grids loaded perpendicularly to the plane of the structure:

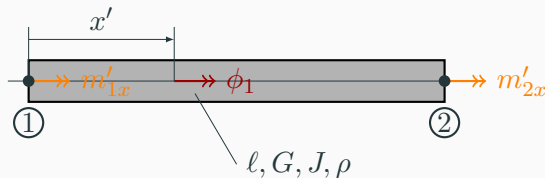
Components required to describe the displacements of joint i are:

- q_{iz} translation in the z direction,
- ϕ_{ix} torsion, rotation about the x axis .
- ϕ_{iy} bending, rotation about the z axis.

Shaft element

The development of a shaft finite element is very similar to the development of a bar finite element, where

- the axial displacement u_1 is replaced by the angular rotation ϕ_1 ,
- the axial nodal forces f'_{ix} are replaced by nodal torque m'_{ix} ,
- the element tensile stiffness AE/ℓ is replaced by the torsional stiffness GJ/ℓ .



- G shear modulus of the material
- J polar moment of inertia of the cross-section.
- ρ material density
- ℓ length
- ϕ_1 angular rotation
- x' (local) axial coordinate

Equation of motion for non-oriented shaft element

Differential equation governing the dynamics:

$$GJ\partial_{x'x'}^2\phi_1(x',t) = \rho J\ddot{\phi}_1(x',t)$$

- Angular rotation (twisting) approximation:

$$\phi_1^h(x',t) = \mathbf{H}(x')\mathbf{q}_{loc}(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{2x}(t) \end{bmatrix}$$

- Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell} \quad \text{and} \quad h_2(x') = \frac{x'}{\ell}$$

- Semi-discrete weak form:

$$\delta\mathbf{q}_{loc}^T(\mathbf{M}_{loc}\ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc}\mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)) = 0$$

Discretization of shaft

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell GJ \frac{d\mathbf{H}^T}{dx'} \frac{d\mathbf{H}}{dx'} dx' = \int_0^\ell GJ \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{GJ}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \int_0^\ell \rho J \mathbf{H}^T \mathbf{H} dx' = \int_0^\ell \rho J \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho J \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

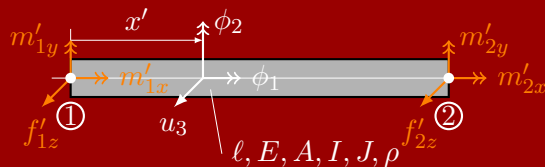
$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} m'_{1x}(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} m'_{2x}(t) = \begin{bmatrix} m'_{1x}(t) \\ m'_{2x}(t) \end{bmatrix}$$

Shaft + Thin beams = Grid



Torsional effects in thin beam element

The inclusion of torsional stiffness in a thin beam element, to model a typical element in a planar grid frame, requires a *superposition* of shaft and beam elements:



Differential equations governing the dynamics:

$$\partial_{x'x'}^2 (EI \partial_{x'x'}^2 u_3(x', t)) + \rho A \ddot{u}_3(x', t) = 0$$

$$GJ \partial_{x'x'}^2 \phi_1(x', t) = \rho J \ddot{\phi}_1(x', t)$$

Torsional effects in thin beam element

Each grid element now has three degrees of freedom per node: ϕ'_{ix} , ϕ'_{iy} and q'_{iz} .

- Displacements approximation:

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t) = [h_1(x') \ h_2(x') \ h_3(x') \ h_4(x') \ h_5(x') \ h_6(x')] \begin{bmatrix} \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ q'_{2z}(t) \end{bmatrix}$$

- Local shape functions:

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = x'(1 - x'/\ell)^2$$

$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_4(x') = x'/\ell$$

$$h_5(x') = x'(x'/\ell)(x'/\ell - 1)$$

$$h_6(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

Discretization of thin beam with torsional effects

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \frac{EI}{\ell^3} \begin{bmatrix} GJ\ell^2/EI & 0 & 0 & -GJ\ell^2/EI & 0 & 0 \\ & 4\ell^2 & -6\ell & 0 & 2\ell^2 & 6\ell \\ & & 12 & 0 & -6\ell & -12 \\ & & & GJ\ell^2/EI & 0 & 0 \\ & & & & 4\ell^2 & 6\ell \\ sym. & & & & & 12 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140J/A & 0 & 0 & 70J/A & 0 & 0 \\ & 0 & 4\ell^2 & 22\ell & 0 & -3\ell^2 & 13\ell \\ & & & 156 & 0 & -13\ell & 54 \\ & & & & 140J/A & 0 & 0 \\ & & & & & 4\ell^2 & -22\ell \\ sym. & & & & & & 156 \end{bmatrix}$$

Arbitrarily oriented thin beam element with torsional effects

- Displacements in local coordinates:

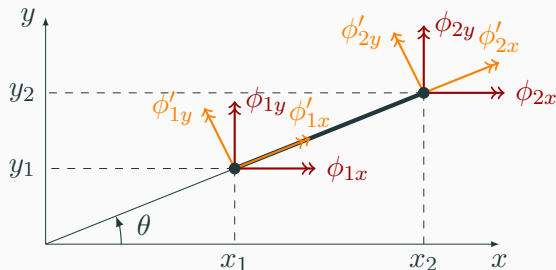
$$\mathbf{q}_{loc} = [\phi'_{1x} \quad \phi'_{1y} \quad q'_{1z} \quad \phi'_{2x} \quad \phi'_{2y} \quad q'_{2z}]^T.$$

- Displacements in global coordinates:

$$\mathbf{q} = [\phi_{1x} \quad \phi_{1y} \quad q_{1z} \quad \phi_{2x} \quad \phi_{2y} \quad q_{2z}]^T.$$

- Relation between local and global displacements:

$$\underbrace{\begin{bmatrix} \phi'_{1x} \\ \phi'_{1y} \\ q'_{1z} \\ \phi'_{2x} \\ \phi'_{2y} \\ q'_{2z} \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 & 0 & 0 \\ -\sin(\theta) & \cos(\theta) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & 0 & 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} \phi_{1x} \\ \phi_{1y} \\ q_{1z} \\ \phi_{2x} \\ \phi_{2y} \\ q_{2z} \end{bmatrix}}_{\mathbf{q}}$$



Discretization of arbitrarily oriented thin beam

- Element stiffness matrix in global coordinates:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T}$$

where $C = \cos(\theta)$ and $S = \sin(\theta)$.

- Element consistent mass matrix in global coordinates:

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T}$$

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc}$$

MATLAB example - dynamic response of a 2d grid

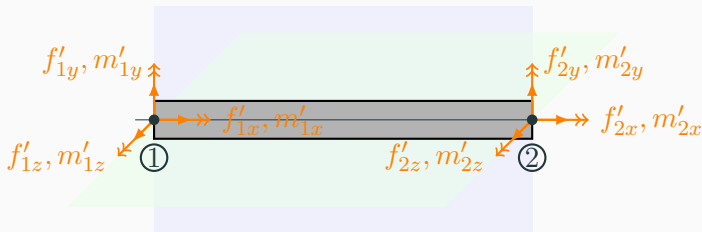
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Three-dimensional frames

Example of a three-dimensional structure composed of thin beams



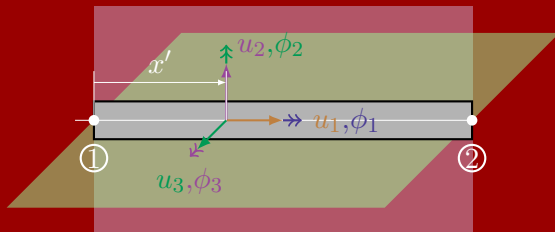
Credit: [N]



Three-dimensional thin beams are uniaxial (slender) element that can support:

- axial loads f'_{ix} ,
- torsional loads m'_{ix} ,
- bending in the $x' - y'$ plane: f'_{iy} and m'_{iz} ,
- bending in the $x' - z'$ plane: f'_{iz} and m'_{iy} .

Differential equations governing the dynamics of thin 3d beams



$$EA\partial_{x'x'}^2 u_1(x', t) = \rho A\ddot{u}_1(x', t)$$

$$GJ\partial_{x'x'}^2 \phi_1(x', t) = \rho J\ddot{\phi}_1(x', t)$$

$$\partial_{x'x'}^2 (EI_z \partial_{x'x'}^2 u_2(x', t)) + \rho A\ddot{u}_2(x', t) = 0$$

$$\partial_{x'x'}^2 (EI_y \partial_{x'x'}^2 u_3(x', t)) + \rho A\ddot{u}_3(x', t) = 0$$

- E Young's modulus of the material.
- A cross-sectional area.
- ρ material density.
- I_y and I_z are the second moments of inertia with respect to Oy and Oz .
- J polar moment of inertia of the cross-section.

Displacements discretization

Total of six nodal generalized displacements at each unconstrained joint:

- three translation components q'_{ix} , q'_{iy} and q'_{iz} along the x , y and z axes,
- three rotational components ϕ'_{ix} , ϕ'_{iy} and ϕ'_{iz} about the x , y and z axes.

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t)$$

$$\mathbf{q}_{loc}(t) = \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_4(x') = 1 - x'/\ell$$

$$h_5(x') = x'(1 - x'/\ell)^2$$

$$h_6(x') = x'(1 - x'/\ell)^2$$

$$h_7(x') = x'/\ell$$

$$h_8(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_9(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_{10}(x') = x'/\ell$$

$$h_{11}(x') = x'(x'/\ell)(x'/\ell - 1)$$

$$h_{12}(x') = x'(x'/\ell)(x'/\ell - 1)$$

Element stiffness matrix in local coordinates

The stiffness matrix for a three-dimensional uniform beam element is written as the superposition of the **axial stiffness** matrix, the **torsional stiffness** matrix and the **flexural stiffness** matrix:

$$\mathbf{K}_{loc} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} \\ 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4EI_z}{\ell} & 0 & -\frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{2EI_z}{\ell} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4EI_y}{\ell} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4EI_z}{\ell} \end{bmatrix}$$

sym.

Element consistent mass matrix in local coordinates

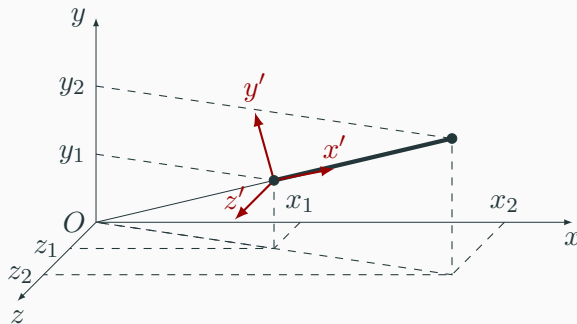
The consistent mass matrix for a three-dimensional uniform beam element is written as the superposition of the **axial mass** matrix, the **torsional mass** matrix and the **flexural mass** matrix:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{420} \begin{bmatrix} 140 & 0 & 0 & 0 & 0 & 0 & 70 & 0 & 0 & 0 & 0 & 0 \\ & 156 & 0 & 0 & 0 & 22\ell & 0 & 54 & 0 & 0 & 0 & -13\ell \\ & & 156 & 0 & -22\ell & 0 & 0 & 0 & 54 & 0 & 13\ell & 0 \\ & & & \frac{140J}{A} & 0 & 0 & 0 & 0 & 0 & \frac{70J}{A} & 0 & 0 \\ & & & & 4\ell^2 & 0 & 0 & 0 & -13\ell & 0 & -3\ell^2 & 0 \\ & & & & & 4\ell^2 & 0 & 13\ell & 0 & 0 & 0 & -3\ell^2 \\ & & & & & & 140 & 0 & 0 & 0 & 0 & 0 \\ & & & & & & & 156 & 0 & 0 & 0 & -22\ell \\ & & & & & & & & 156 & 0 & 22\ell & 0 \\ & & & & & & & & & \frac{140J}{A} & 0 & 0 \\ & & & & & & & & & & 4\ell^2 & 0 \\ & & & & & & & & & & & 4\ell^2 \end{bmatrix}$$

sym.

Transformation of coordinates

- The stiffness and mass matrices are defined in the local coordinate system x' , y' and z' fixed to the beam segment.
- To assemble the global stiffness and mass matrices, these local matrices must be transformed into the global coordinate system x , y , z .



Direction cosines

- The relationship between local and global coordinates is:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos(x'x) & \cos(x'y) & \cos(x'z) \\ \cos(y'x) & \cos(y'y) & \cos(y'z) \\ \cos(z'x) & \cos(z'y) & \cos(z'z) \end{bmatrix}}_{\mathbf{T}'} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- The local x' is given by:

$$Ox' = [l_x, l_y, l_z]$$

where

$$l_x = \cos(x'x) = \frac{x_2 - x_1}{e_\ell}, \quad l_y = \cos(x'y) = \frac{y_2 - y_1}{e_\ell}, \quad l_z = \cos(x'z) = \frac{z_2 - z_1}{e_\ell},$$

$$e_\ell = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Direction cosines

- The local y' axis is chosen so that it lies in the local $x' - y'$ plane and it is perpendicular to x' :

$$Oy' = \frac{1}{d} [l_y, l_x, 0]$$

thus

$$\cos(y'x) = \frac{l_y}{d}, \quad \cos(y'y) = \frac{l_x}{d}, \quad \cos(y'z) = 0, \quad d = \sqrt{l_x^2 + l_y^2}.$$

- The local z' axis is chosen so that $Oz' = Ox' \times Oy'$:

$$Oz' = \frac{1}{d} [-l_x l_z, -l_y l_z, 0]$$

thus

$$\cos(z'x) = -\frac{l_x l_z}{d}, \quad \cos(z'y) = -\frac{l_y l_z}{d}, \quad \cos(z'z) = d.$$

Global displacements for oriented three-dimensional beam

$$\underbrace{\begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}}_{\mathbf{q}_{loc}(t)} = \underbrace{\begin{bmatrix} \mathbf{T}' & & \\ & \mathbf{T}' & \\ & & \mathbf{T}' \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x}(t) \\ q_{1y}(t) \\ q_{1z}(t) \\ \phi_{1x}(t) \\ \phi_{1y}(t) \\ \phi_{1z}(t) \\ q_{2x}(t) \\ q_{2y}(t) \\ q_{2z}(t) \\ \phi_{2x}(t) \\ \phi_{2y}(t) \\ \phi_{2z}(t) \end{bmatrix}}_{\mathbf{q}(t)}$$

